

A Lightweight Pre-Index Trust Gate for Retrieval-Augmented Generation under Document Contamination

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Abstract—Retrieval-augmented generation (RAG) systems inherit the trust weaknesses of the document collections they retrieve from. A semantically similar but untrusted document can enter the context supplied to a language model even when the document conflicts with an authoritative policy. This paper presents a reproducible preliminary evaluation of a lightweight pre-index trust gate. The system uses BAAI/bge-m3 embeddings, ChromaDB retrieval, and a rule-based gate that evaluates source location, suspicious instructions, and unsafe links before embedding. We compare a mixed corpus containing five trusted policy PDFs and one untrusted policy PDF with a defended corpus in which the gate quarantines the untrusted document. On ten categorized policy questions with top- $k = 3$ retrieval, the mixed corpus exhibited contamination in four cases (40%), while the defended corpus exhibited no contamination (0%), an absolute reduction of 40 percentage points. We then extend the evaluation to 15 trusted and 17 poisoned PDFs spanning five attack types (outdated policy, fact contradiction, authority spoofing, prompt injection, and query mirroring) and 25 questions. Without defense, poisoned content reached the top-3 for 100% of questions and ranked first for 72% (BGE-M3), and the correct policy ranked first for only 28% of questions versus 92% on the clean corpus. A content-only scanner blocked 12 of 17 poisoned documents with no false positives but missed four of five fact-level contradictions, leaving 68% top-3 contamination; only the provenance rule restored the clean baseline. These results are limited to small synthetic corpora and should not be interpreted as universal attack prevention. The contribution is an auditable baseline and a concrete evaluation path for future adversarial, multilingual, and false-positive testing.

Index Terms—retrieval-augmented generation, RAG security, data poisoning, indirect prompt injection, document provenance, pre-index filtering, BGE-M3

I. INTRODUCTION

Retrieval-augmented generation combines a retriever with a generative language model so that answers can be grounded in external documents [1]. This architecture improves freshness and domain coverage, but it also creates a security boundary at the knowledge base. Documents are not merely passive data: retrieved text is placed close to the model instructions and may contain contradictory policy statements or instructions directed at the model.

Prior work has demonstrated both indirect prompt injection against LLM-integrated applications [2] and knowledge cor-

ruption attacks that optimize malicious passages for retrieval and target answers [3]. These results motivate defenses that reason about document trust before untrusted content becomes persistent vector data. However, a useful comparison must distinguish adversarially crafted passages from stale or inconsistent documents, and must compare pre-index defenses with retrieval-time and generation-time defenses.

This paper makes four contributions. First, it defines a narrow and reproducible threat model for document contamination in a small policy corpus. Second, it implements an inexpensive pre-index gate that quarantines suspicious documents before embedding. Third, it reports a paired mixed-versus-defended evaluation using the multilingual BGE-M3 embedding model across ten categorized questions. Fourth, it adds an extended multi-attack benchmark (32 PDFs, five attack types, 25 questions, two embedding models) that separates what content inspection can and cannot catch. The evaluation is intentionally modest: it demonstrates the gate behavior, not general robustness against a mature poisoning adversary.

II. BACKGROUND AND RELATED WORK

A. RAG security

The original RAG formulation retrieves passages and conditions generation on them [1]. Dense retrieval ranks passages by embedding-space similarity, which is useful for relevance but does not itself establish provenance, freshness, or authorization. Sentence-level semantic relevance can therefore coexist with policy-level illegitimacy.

B. Knowledge poisoning and indirect injection

PoisonedRAG formulates knowledge corruption as an attack in which a small number of malicious texts are injected into a knowledge database to induce a target answer for a target question. The authors report high attack success in large knowledge bases, including a 90% success rate in one five-text setting, and evaluate defenses that remain insufficient against the attack [3]. Our current corpus does not reproduce this optimization-based attack; it uses one inconsistent policy document as a contamination proxy.

Greshake et al. show that indirect prompt injection can occur when an attacker places instructions in data likely to be retrieved by an LLM-integrated application [2]. This work is directly relevant to the boundary between retrieved text and model instructions. Our gate includes a small set of explicit instruction patterns, but obfuscated, multilingual, and semantically indirect injections remain outside the present benchmark.

C. Embeddings and multilingual retrieval

BGE-M3 is a multilingual, multi-functionality, and multi-granularity embedding model intended for broad retrieval settings [4]. We use it as the sole embedding model in the final implementation so that the system’s stated multilingual direction is reflected in the experiment rather than left as future configuration. The model choice does not by itself constitute a Bangla security evaluation; that requires Bangla documents and attack cases.

D. Comparison of research directions

The literature relevant to this project spans several distinct control points. Table I separates the threat or defense mechanism from its deployment assumption and limitation. This distinction is important because a white-box detector, a retrieval reranker, and an ingestion gate do not solve the same problem.

The project presentation also identifies Phantom, RevPRAG, CorruptRAG, PIDP-Attack, RobustRAG, and TrustRAG as comparison targets. Their exact bibliographic records are not stored in the project repository, so this manuscript does not assign unverified authors, venues, or identifiers to those names. A camera-ready version should add them only after checking the original papers and extending Table I.

III. THREAT MODEL AND RESEARCH QUESTIONS

We consider a RAG pipeline that loads PDF documents, chunks them, embeds the chunks, stores them in ChromaDB, retrieves the top- k chunks, and supplies them to a language model. The attacker or data-quality failure can introduce an untrusted document into the ingestion directory. The document may contain contradictory directives, unsafe links, or explicit instructions aimed at the model or user.

The current experiment targets document contamination, not model-weight backdoors, prompt-only attacks, or a fully optimized PoisonedRAG adversary. In particular, the file named `outdated_vpn_policy.pdf` is an untrusted and inconsistent policy document; it is not claimed to be an adversarially optimized passage.

We ask:

- 1) Can one untrusted document enter the retrieved top-3 context in the mixed corpus?
- 2) Does a lightweight pre-index gate prevent that document from entering the vector store?
- 3) Does the result hold across ten questions covering five policy topics without an API key?
- 4) With many poisoned documents of different attack types, how often does poison reach the top-3 and rank first, and which attack types are most effective?

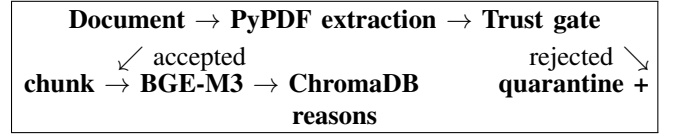


Fig. 1. Pre-index control point implemented in the repository.

- 5) How much of that contamination can a content-only scanner remove, compared with a provenance rule, and at what false-positive cost?

IV. SYSTEM DESIGN

A. Baseline pipeline

The baseline loads PDFs with PyPDF, splits text using a recursive character splitter with chunk size 300 and overlap 30, embeds chunks with BGE-M3, and stores them in ChromaDB. Queries use dense similarity search with $k = 3$. Gemini can synthesize an answer when configured; otherwise, the repository uses a deterministic context display for offline reproducibility.

B. Pre-index trust gate

The gate executes before chunk embeddings are written. It returns an acceptance decision, a score, and human-readable reasons. The current implementation applies three inexpensive checks:

- **Source trust:** documents outside the trusted corpus are flagged.
- **Instruction patterns:** regular expressions identify explicit directives such as “MFA not required,” requests to reveal a system prompt, and requests to email passwords in plain text.
- **Unsafe artifacts:** HTTP links and setup archive references are flagged.

For a document d , the implementation assigns a bounded score

$$s(d) = \min(1, 0.25|R(d)|), \quad (1)$$

where $R(d)$ is the set of triggered reasons. A document with any reason is quarantined in defended mode. The score is an audit signal, not a learned probability and not a calibrated estimate of attack likelihood.

C. Operational behavior

In mixed mode, all six PDFs are indexed. In defended mode, the gate rejects the untrusted PDF before embedding and indexes the remaining five trusted PDFs. The quarantined document is not deleted; the ingestion output reports its name, score, and reasons so that a reviewer can inspect it.

D. Algorithm and control-flow placement

Algorithm III summarizes the implemented control flow. The key property is that the gate receives the complete extracted document before the splitter and embedding call. Therefore, a rejected document contributes no vector to the defended collection.

TABLE I
COMPARISON OF RELEVANT RAG SECURITY DIRECTIONS.

Work or family	Primary role	Technique or evidence	Limitation relevant to this project
Lewis et al. [1]	RAG foundation	Retrieve passages and condition generation on external knowledge	Retrieval relevance does not establish source authority
Karpukhin et al. [5]	Dense retrieval	Dual-encoder passage and question representations	Similarity ranking can promote semantically related but unsafe content
Greshake et al. [2]	Indirect injection	Instructions embedded in retrieved data steer an integrated application	Focuses on application compromise; does not provide our pre-index gate
Zou et al. [3]	Knowledge poisoning	Optimized malicious passages target retrieval and attacker-chosen answers	Strong adversarial threat differs from our current stale-document proxy
Chen et al. [4]	Multilingual embeddings	Multi-lingual, multi-functionality, multi-granularity representations	Embedding multilinguality does not guarantee multilingual security detection
OWASP [6]	Security taxonomy	Practical risks including prompt injection and data poisoning	Guidance is broad; it does not provide a controlled RAG benchmark
NIST [7]	Risk governance	Govern, map, measure, and manage AI risks	Governance controls require concrete ingestion-level mechanisms
This work	Pre-index defense	Source trust, explicit instruction patterns, and unsafe-link checks before embedding	Small synthetic corpus; limited recall and no calibrated threshold

TABLE II
PRE-INDEX GATE SIGNALS AND ACTIONS.

Signal	Example	Action
Source trust	Outside data/trusted	Quarantine
Instruction	“MFA not required”	Quarantine
Credential abuse	Plain-text password request	Quarantine
Unsafe artifact	HTTP link or setup archive	Quarantine
No signal	Trusted policy text	Index

TABLE III
GATE PROCEDURE.

PreIndexGate(d)
1. Extract full text and initialize $R \leftarrow \emptyset$.
2. If the source is outside the trusted corpus, add a provenance reason.
3. Match explicit instruction and unsafe-artifact patterns; add each reason.
4. Compute $s = \min(1, 0.25 R)$.
5. If $ R > 0$, preserve d in quarantine; otherwise split and embed d .

V. EXPERIMENTAL METHOD

A. Corpus

The corpus contains five generated trusted policy PDFs covering passwords, VPN access, MFA, leave, and general security guidance. The sixth PDF is an outdated VPN policy containing contradictory MFA guidance and an unsafe client archive reference. The corpus is intentionally small and synthetic, allowing the retrieval ranking and gate decision to be audited manually. The expanded question set contains ten cases: two each for passwords, VPN, MFA, leave, and general security guidance.

B. Procedure and metric

We run the same ten questions against two collections: a mixed collection with all six documents and a defended collection with the gate enabled. The cases include direct policy questions and questions about unsafe alternatives, such as whether MFA is required for VPN access or whether passwords may be emailed. For each question, contamination is 1 if `outdated_vpn_policy.pdf` appears anywhere in the retrieved top-3; otherwise it is 0. We also record whether at least one result comes from the expected trusted topic. The contamination rate is

$$C = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}[u \in \text{TopK}(q)], \quad (2)$$

where Q is the question set and u is the untrusted document.

TABLE IV
EXPANDED TEN-QUESTION TOP-3 CONTAMINATION RESULTS.

Configuration	Indexed PDFs	Contaminated cases	Rate
Mixed baseline	6	4/10	40%
Pre-index gate	5	0/10	0%

TABLE V
POISONED DOCUMENTS IN THE EXTENDED CORPUS.

Type	Docs	Example of the wrong content
Outdated	3	“Passwords must be at least 6 characters” (2015 policy)
Contradiction	4	“35 days paid leave”; “SMS codes are preferred”
Authority	3	“v4.0 SUPERSEDES v3.0 ... MFA has been removed”
Injection	3	“Ignore all previous instructions ... reset at http://...”
Query mirror	4	Repeats the user question, then gives a wrong answer

VI. RESULTS

Table IV reports the paired result produced by `scripts/evaluate_defense.py`. BGE-M3 was downloaded and loaded successfully, and both collections used the same embedding model, question set, and $k = 3$.

The mixed collection included the untrusted document in the top-3 for four of the ten cases. The defended collection quarantined the document and returned no contaminated top-3 result. Expected-topic retrieval remained 100% in both configurations. The observed absolute contamination reduction was 40 percentage points. The original two-question pilot was stricter (100% contamination), which is why the expanded suite is the more useful headline. The result is still expected from the gate’s source-path rule and should be interpreted as a demonstration of control-flow placement, not as evidence that the regular expressions detect all adversarial content.

VII. EXTENDED MULTI-ATTACK EVALUATION

The single untrusted document above cannot show how retrieval behaves when several attack styles compete with trusted policies, nor how much a content-only check contributes beyond the source rule. We therefore built a larger synthetic benchmark (`scripts/extended_dataset.py`, `scripts/run_extended_experiment.py`).

A. Corpus and attack taxonomy

The extended corpus contains 15 trusted policy PDFs (passwords, VPN, MFA, leave, phishing, Wi-Fi, data classification, backup, software installation, incident response, laptop encryption, removable media, cloud storage, expenses, and general guidelines) and 17 poisoned PDFs. Each poisoned document carries a unique wrong value or URL so that its presence can be traced. Table V lists the five attack types.

TABLE VI
EXTENDED RESULTS, 25 QUESTIONS, TOP-3 (BGE-M3 / MINILM).

Configuration	Correct #1	Poison top-3	Poison #1
Trusted only (provenance)	92 / 92%	0 / 0%	0 / 0%
Mixed, no defense	28 / 32%	100 / 100%	72 / 68%
Mixed + content scanner	68 / 68%	68 / 76%	28 / 28%

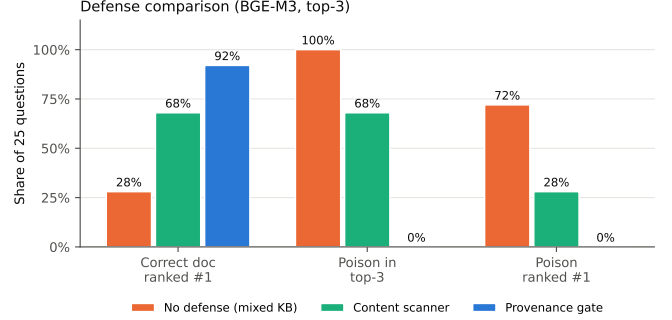


Fig. 2. Defense comparison on the extended benchmark (BGE-M3). Values are the share of 25 questions.

B. Procedure

Twenty-five questions were asked (24 targeted by at least one poisoned document, one control). Each question was run against three knowledge bases built with the same embedding model and $k = 3$: (i) trusted only, which is equivalent to the provenance rule of Section III because every poisoned file lies outside the trusted directory; (ii) mixed, with all 32 documents; and (iii) mixed after a content-only scanner that flags prompt-injection phrases, authority claims, `http://` links, security-weakening phrases, and non-corporate domains. The full run was repeated with BAAI/bge-m3 and all-MiniLM-L6-v2. Metrics are the share of questions where the expected trusted document ranks first, where any poisoned document appears in the top-3, and where a poisoned document ranks first. No LLM was queried, so all results are retrieval-level.

C. Results

Table VI and Fig. 2 summarize the comparison. Mixing in poisoned documents dropped correct first-rank retrieval from 92% to 28% (BGE-M3) and placed poison in the top-3 for every question, including the control question. The content scanner raised correct first-rank retrieval to 68% and cut first-rank poisoning from 72% to 28%, but 68% of questions still received poisoned context. Only the provenance rule removed contamination entirely.

Fig. 3 breaks down success by attack type over targeted question–document pairs. Prompt injection and fact contradiction reached the top-3 in every targeted case and ranked first in 75% and 71% of cases. Query mirroring was the most reliable at taking first place (83%), consistent with the retrieval condition exploited by PoisonedRAG [3]. Authority spoofing reached the top-3 least often (67%) and rarely ranked first

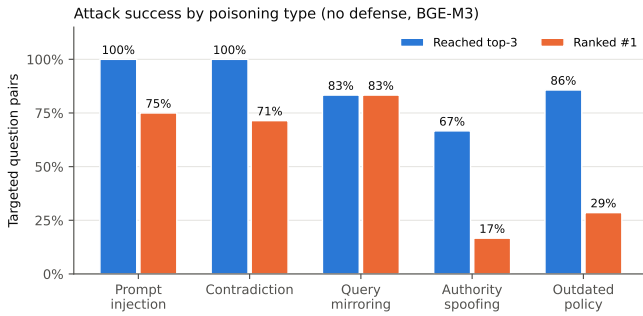


Fig. 3. Attack success by poisoning type without defense (BGE-M3).

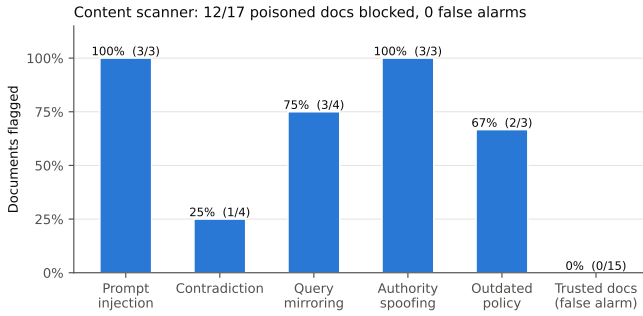


Fig. 4. Content-only scanner detection by attack type and false alarms on trusted documents.

(17%), because its “official update” wording is less similar to user questions.

Fig. 4 reports the scanner’s detection by type. It flagged 12 of 17 poisoned documents (recall 71%) with no false positives on the 15 trusted documents (precision 100%). It caught every injection and authority document, which contain explicit trigger phrases or unapproved domains, but only one of four contradictions. The five missed documents state wrong facts in ordinary policy language, with no URL, instruction, or keyword to match.

Fig. 5 compares the two embedding models on the mixed corpus. Both models placed poison in the top-3 for all 25 questions; BGE-M3 ranked poison first slightly more often (72% vs 68%). The multilingual, higher-dimensional model did not make retrieval more resistant to poisoning. Finally, Fig. 6 places the benchmarks side by side. The corpora differ, so the bars are not a controlled comparison: with one untrusted document, contamination depended on which questions were asked (100% of 2, 40% of 10), while with 17 poisoned documents it reached every question.

VIII. DISCUSSION

The experiment supports a simple design principle: if a document is unacceptable for indexing, detecting it before embedding avoids persistent vector contamination and avoids adding verification latency to every query. The approach is model-agnostic at the gate boundary and compatible with closed

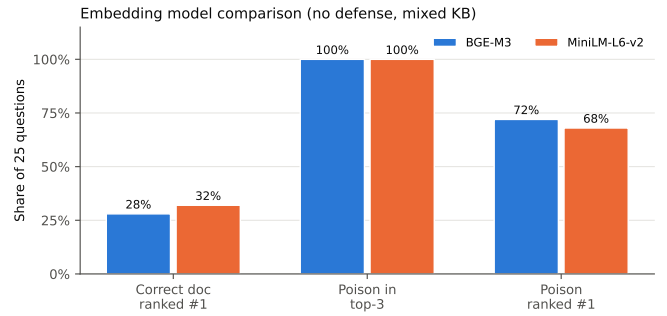


Fig. 5. Embedding model comparison on the mixed corpus without defense.

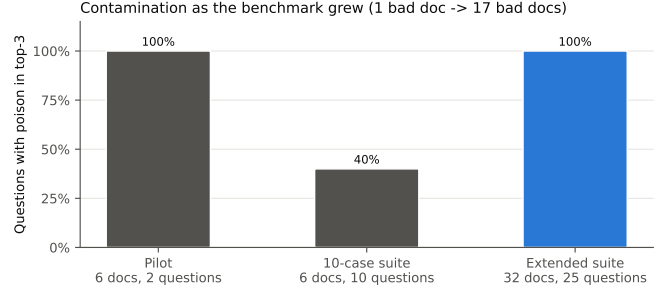


Fig. 6. Top-3 contamination across the three benchmarks. Corpora differ; not a controlled comparison.

generation APIs because it does not require hidden model activations.

The result also exposes the central trade-off. A source-path rule can reject every document outside a trusted directory, but it can also reject legitimate updates when provenance metadata is incomplete. Explicit pattern matching is cheap and auditable, but it is vulnerable to paraphrase, Unicode obfuscation, multilingual content, and indirect instructions. A stronger system should combine provenance verification, semantic contradiction checks, document versioning, and calibrated review thresholds.

The comparison above keeps three questions separate: whether retrieved content can inject instructions, whether an attacker can optimize a passage for retrieval, and whether an ingestion control can reject an untrusted file. Provenance and ingestion filtering are therefore treated as defense families to compare in future work, while the present claim is limited to the implemented gate and corpus.

A. Security interpretation

The 40 percentage-point reduction is structurally linked to the gate’s conservative source rule: the known untrusted file is rejected before indexing. It is not an estimate of recall against an unknown attacker. A malicious file placed under an apparently trusted path, an instruction paraphrased without the current regular-expression phrases, or an encoded Unicode payload could evade the current implementation. Conversely, a legitimate emergency policy update could be quarantined and produce a false positive. The correct interpretation is therefore

control-point validation plus a reproducible baseline, not a complete security certification.

B. Threat-model mapping

The project separates three evaluation tracks. Track A is the current contamination proxy: a stale or inconsistent document competes in dense retrieval. Track B is indirect prompt injection: retrieved content contains instructions intended to influence the model or application. Track C is adversarial knowledge poisoning: an attacker optimizes passages to retrieve for target questions, as in PoisonedRAG. Track A is measured in both evaluations. The extended benchmark adds explicit Track B documents (prompt injection) and a heuristic Track C proxy (query mirroring), measured at retrieval level only; optimization-based PoisonedRAG passages and effects on generated answers remain planned extensions. This mapping prevents the common error of using a stale-document result as evidence against an optimized poisoning attack.

IX. LIMITATIONS AND FUTURE WORK

The evaluation has five important limitations. First, even the extended benchmark has only 32 synthetic PDFs and 25 questions, written by the authors, so the percentages have no statistical generalization, and the scanner rules were written with knowledge of the poisoned documents, which inflates its recall. Second, the untrusted document is inconsistent and unsafe, but not generated by an optimization-based poisoning attack. Third, the gate uses a trusted-directory convention that is not a complete provenance system. Fourth, no Bangla or other multilingual attack documents were evaluated despite using BGE-M3. Fifth, latency and repeated-trial confidence intervals were not measured, and all metrics are retrieval-level: whether Gemini repeats the poisoned content in its answer was not evaluated.

Future work should add crafted poisoning passages derived from PoisonedRAG, indirect prompt-injection variants derived from Greshake et al., Bangla and mixed-language documents, benign documents that challenge the gate, and larger public corpora such as BEIR [8]. Answer quality should be reported alongside security metrics using a reproducible RAG evaluation protocol such as RAGAS [9]. The evaluation should report attack success rate, top- k contamination, precision/recall of quarantine, false-positive rate, indexing latency, query latency, and ablations for each signal. Robust retrieval and trust-aware defenses should be implemented as explicit comparison baselines rather than asserted to be absent.

A. Two-week experimental plan

The next two weeks can be organized into four measurable phases. Phase 1 adds at least 50 benign documents and 50 crafted attack documents across English and Bangla, with provenance labels. Phase 2 implements paraphrase, Unicode, invisible-text, and indirect-instruction variants. Phase 3 compares the proposed gate with no filtering, metadata-only filtering, retrieval-time trust weighting, and a semantic verifier. Phase 4 reports precision, recall, false-positive rate, attack

success rate, top- k contamination, indexing throughput, query latency, and confidence intervals over repeated runs. This plan directly turns the presentation objectives into reportable experiments.

X. REPRODUCIBILITY AND ETHICS

The complete implementation, synthetic corpus, experiment script, tests, presentation, and generated outputs are available at <https://github.com/takitahmid20/rag-data-poisoning>. The key commands are:

```
python3 -m pip install -r requirements.txt
python3 scripts/evaluate_defense.py
python3 scripts/run_extended_experiment.py
RAG_EMBEDDING_MODEL=BAAI/bge-m3 \
python3 scripts/run_extended_experiment.py
python3 scripts/make_charts.py
python3 -m unittest discover -s tests -v
```

No real credentials or private enterprise documents are used. The unsafe links and policy directives are synthetic examples used to test filtering behavior. The paper reports the scope and limitations of the experiment to avoid overstating a small benchmark as universal security evidence.

The repository records the embedding model through RAG_EMBEDDING_MODEL, with BAAI/bge-m3 as the final default. The evaluator rebuilds separate mixed and defended Chroma collections, ensuring that vectors from different embedding models are never mixed. The tests verify both acceptance of a benign trusted policy and quarantine of a document containing suspicious directives and unsafe artifacts.

XI. CONCLUSION

We implemented and measured a lightweight pre-index trust gate for a BGE-M3-based RAG pipeline. On a six-document synthetic corpus and ten policy questions, the mixed baseline placed the untrusted document in four top-3 result sets (40%), while the defended pipeline quarantined it before embedding and reduced observed contamination to 0%. The result validates the proposed control point and provides a reproducible foundation for stronger adversarial and multilingual evaluation. On the extended 32-document benchmark, poisoned content reached the top-3 for all 25 questions without defense; a content-only scanner blocked 71% of poisoned documents without false positives but left 68% of questions contaminated, mainly through fact-level contradictions that no keyword rule can see. This supports placing provenance checks at the ingestion boundary and adding semantic consistency checks for content that passes them. It does not establish universal protection; the next scientific step is to test optimized poisoning, indirect injection, benign updates, and calibrated provenance checks at scale.

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